PROJECT PROPOSAL

AI-Assisted Maze Navigation with N-Puzzle Challenge

A Hybrid Intelligent System for Algorithmic Problem Solving

PROJECT CREDITS

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PROJECT DETAILS

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STATUS: Development Phase

01. EXECUTIVE OVERVIEW

Interactive games are the ultimate frontier for testing Artificial Intelligence. This project introduces a **dual-layered AI framework** where navigation and logic puzzles are inextricably linked.

Unlike traditional games that either provide total solutions or zero help, this system implements **Conditional Intelligence**: the AI only assists those who prove their logical merit through an N-Puzzle challenge.

02. THE PROBLEM SPACE

The Gap in Game AI

Static Guidance

Most games provide a "line on the floor" that removes all challenge.

Frustration Peaks

Without guidance, players often abandon complex mazes.

The Solution

A gated assistance mechanism where AI reveals only the **next correct direction** at decision points.

03. TECHNICAL

The system is built on a modular, five-pillar architecture designed for high performance and clean separation of concerns.

Module	Purpose	Algorithm
Maze Engine	Grid & Visibility	Matrix Representation
Maze AI	Shortest Path	BFS / A*
N-Puzzle	Sliding Logic	State-Space Modeling
Puzzle AI	Move Prediction	Manhattan Heuristic
GUI Layer	User Interface	Pygame / Modern UI

04. CORE METHODOLOGY

I. Maze Navigation (The Macro AI) : A*

The AI calculates the global shortest path from the player's current *(x, y)* to

- **Heuristic Function:**

$$f(n) = g(n) + h(n)$$

Where *h(n)* is the sum of **Manhattan Distances** of all tiles from their goal

II. N-Puzzle Solver (The Micro AI)

To unlock the Maze hint, the player must solve a 3×3 sliding puzzle.

- **Heuristic Function:**

$$f(n) = g(n) + h(n)$$

Where *h(n)* is the sum of **Manhattan Distances** of all tiles from their goal

05. UNIQUE INNOVATIONS

Dual AI Collision

Merging pathfinding with heuristic search in a single loop.

Information Gating

The AI acts as a reward, not a shortcut.

Limited Visibility

Enhances the psychological need for the AI "compass."

06. PROJECT ROADMAP

Phase 1
Core Maze & Player Movement.

Phase 2
Implementation of BFS/A* Pathfinding.

Phase 3
N-Puzzle Integration & Manhattan Heuristics.

Phase 4
GUI Polishing & UI/UX Feedback Loops.

07. EXPECTED OUTCOMES

1 Fully Functional Game

A seamless transition between Maze and Puzzle screens.

2 Modular Codebase

Easy to adapt for 4×4 puzzles or larger mazes.

3 Educational Impact

A clear demonstration of how search algorithms work in real-time.